

1D Random Walk

Introduction

A random walk is a **stochastic process** in which a walker — such as a particle or a person — follows a path consisting of a succession of random steps. A classic related example is the **Brownian motion** of particles suspended in a fluid, resulting from collisions with fast-moving molecules in the surrounding medium.

In **statistical physics**, the random walk helps predict:

- The rate at which a gas diffuses through another gas.
- How heat spreads through a solid.
- Fluctuations in pressure inside a small heat bath.

Central Limit Theorem (CLT)

Let $X_1, X_2, \dots, X_n$ be independent, identically distributed random variables in $\mathbb{R}$ with mean 0 and variance $\sigma^2 < \infty$. Then the normalized sum

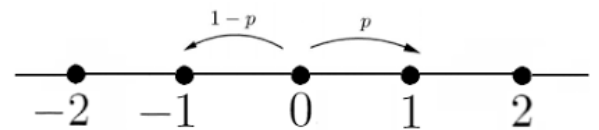
$$Y = \frac{X_1 + \dots + X_n}{\sqrt{n}}$$

converges in distribution to a normal distribution with mean 0 and variance σ^2 as $n \rightarrow \infty$.

Simulation

The one-dimensional random walk is constructed as follows:

1. Initialize the walker at a position $y_0 = 0$
2. Flip a fair coin ($p = 1/2$):
 - a. if it's heads, take one step right
 - b. if it's tails, take one step left
3. Repeat this for N steps



Steps should be of size 1. If u_i is the i^{th} step, then the trajectory up to the n^{th} step is

$$y_n = \sum_{i=1}^n u_i$$

Tasks

1. Plot the trajectory a single random walker versus the number of steps for $N = 500$. Make sure your plot has appropriately labeled axes and a title.

Q1. What is the probability distribution of a single step u_i ? Draw it.

2. Plot an ensemble of 1000 trajectories for $N = 400$ on the same axes. Adjust the `alpha` of your plot to improve readability.

Q2. What is the expected value of a single step $\langle u_i \rangle$? Use this to calculate the mean final position $\langle y_N \rangle$ of a random walker.

Optional Calculate the mean final position if you have a biased coin ($p \neq \frac{1}{2}$)

3. Calculate the standard deviation of your ensemble for each step. Plot the standard deviation and its reflection across the x-axis on top of your previous plot. Make sure to include a legend.

Q3. How does the standard deviation $\sigma_N = \sqrt{\langle y_N^2 \rangle_c}$ depend on N ? In particular, given that

$$\sigma_N \sim N^\alpha$$

What is α ? (hint: look up the `scikit.optimize` module for a way to fit the curve)

Q4. Find the value of α mathematically.

4. Plot your numerical data and the theoretical prediction of the standard deviation on the same plot.

Optional Determine mathematically what the variance and α should be for a biased coin. Does it depend on p ? Interpret the result for $p=1$.

5. Plot a histogram of the final positions of 50,000 random walkers for $N = 10,000$ steps.

Q5. *What is the shape of the histogram? Does it resemble the shape of the probability distribution of a single step you drew in Q1? Why or why not?*

6. Use the `density` keyword to normalize the above histogram into probabilities. Then, import `norm` from the module `scipy.stats` to create a Gaussian fit of your data. Plot this fit on top of your histogram. (*hint: look into the `fit` and `pdf` methods of the `norm` class*)

Q6. *What do you expect the histogram of final positions to look like for a biased coin? Does the central limit theorem apply?*

Optional *Simulate a biased random walk and complete the above activities for different values of p . What is different? What is the same?*